

Subset Selection for Speedy Mass Testing using Graph Signal Processing

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Contents

1	Introduction	4
1.1	Subset Selection Problem	4
1.2	Dynamics of Epidemiology in an Epidemic	5
1.3	BMC - Measles Dataset	6
2	Graph Signal Processing	8
2.1	Graph Signal Processing	8
2.2	Graph Spectral Domain	8
2.3	Graph Laplacian	9
2.4	Compressed Sensing Theory for Reconstruction of Partially known Signals	9
3	Graph Signal Processing for Testing	11
3.1	Graph Frequency Analysis of Covid-19 Incidence to Identify County level Contagion Patterns In the United States	11
3.1.1	GSP fundamentals	12
3.1.2	GFT and Smoothness	12
3.1.3	GSP and COVID-19	13
3.2	Kemeny-based testing for COVID-19	15
4	Graph Signal Processing for Subset Selection with Contact Tracing Graph	17
4.1	Nomenclature And Algorithm	17
5	Results Obtained	22
6	Investigating Graph	24
6.1	Analysis of Results Obtained	24
6.1.1	Necessity of Introducing a Community Structure	24
6.1.2	Subset Selection Strategies	24

List of Figures

1.1	Picking Up the “Right” Subset	4
1.2	The <i>Testing</i> “PARADOX”	5
1.3	Age Wise Statistics	6
1.4	Ward Wise Statistics	6
1.5	Hospital Wise Statistics	7
4.1	Nomenclature Part 1	17
4.2	Nomenclature Part 2	18
4.3	Nomenclature Part 3	18
4.4	Algorithm Part 1	19
4.5	Algorithm Part 2	20
4.6	Algorithm Part 3	21
4.7	Algorithm Part 4	21
5.1	RMSE vs Rho and Tau	22
5.2	RMSE vs Day	22
5.3	Sensitivity vs Day	23
5.4	Specificity vs Day	23
5.5	Theta plot	23

Chapter 1

Introduction

1.1 Subset Selection Problem

Selection of the *right* subset, traditionally enjoys the legacy of being one of the most sought-after problem. Thanks to the ever-expanding world, which makes this problem never lose its charm. Choosing the best representative from a large domain is a problem that fundamentally being ubiquitous and NP-Hard at the same time, suggests always in all contexts where this problem arises, that there is a pressing need for optimizing the procedure of selecting the right subset. The problem has seen many applications to it occurring almost every possible walks of life, be it from picking up the right subset of points to pick from the set of points, to appoint as the cluster heads, and then dispersing off the rest of the points to those zonal heads, which could be commonly called as clustering or picking up the right representative subset from the population which will help identify the full population dynamics. All in all this problem of Subset selection emphasizes strongly an optimization problem, which wants to maximize a function from the minimum possible observatory or scientific effort.

CHOOSING THE CLUSTERING “HEADS” TO BE THE REPRESENTATIVES

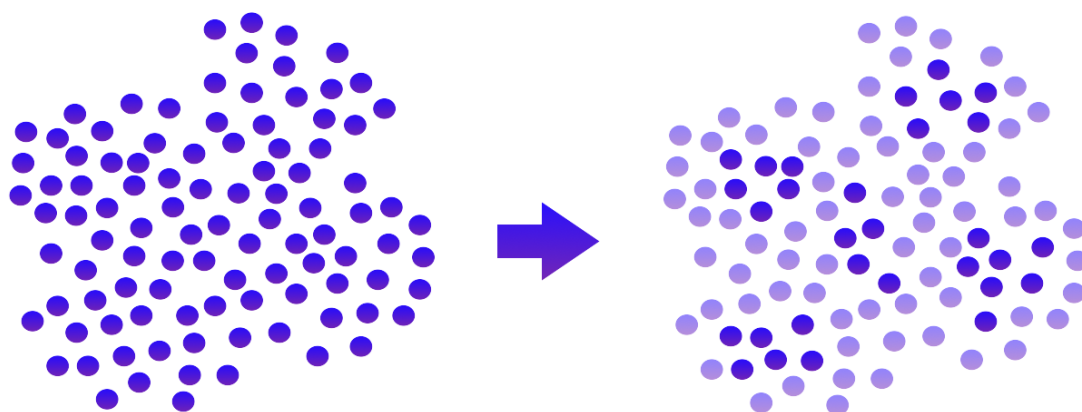


Figure 1.1: Picking Up the “Right” Subset

Presented here is yet another application of Subset selection in the context of Speedy group testing on the verge of an Epidemic. An interesting primer on epidemics and epidemiology is presented below: An epidemic is like a sudden storm

of sickness that sweeps through a community, leaving a trail of coughs, fevers, and aches in its wake. It's like a wildfire of infection, spreading rapidly and threatening to overwhelm the healthcare system. It's a wake-up call that reminds us of the power of infectious diseases and the importance of public health measures to keep them at bay. Epidemiology is like a detective story, but instead of solving crimes, the goal is to unravel the mysteries of diseases and outbreaks. It's a field of study that combines the power of science and data to understand how diseases spread, who is most vulnerable, and how to prevent and control them. Epidemiologists are like disease detectives, using their skills to track down clues and patterns, piece together evidence, and make sense of complex health issues. With epidemiology on our side, we can uncover the secrets of diseases and protect our communities from their deadly grip.

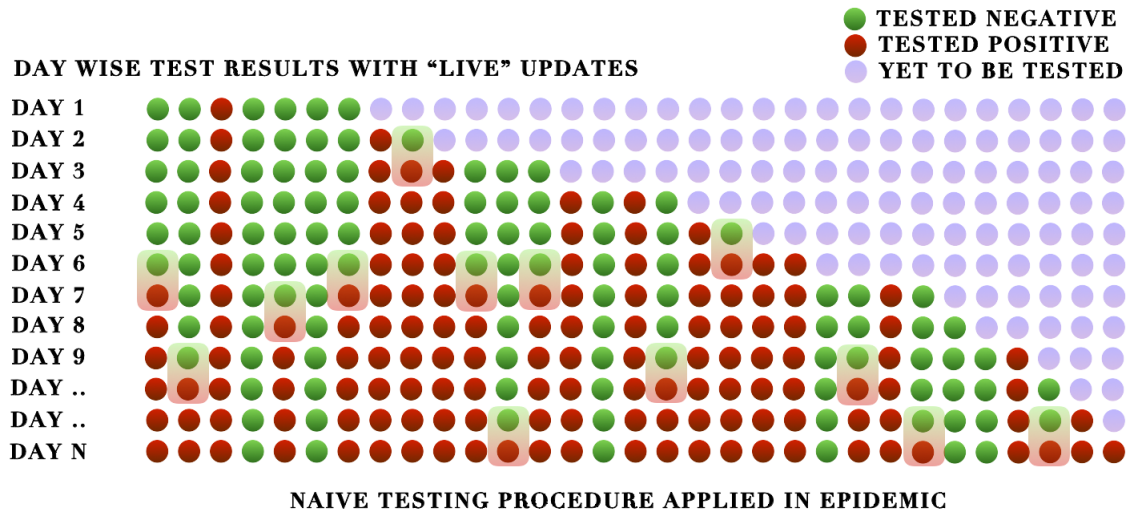


Figure 1.2: The *Testing* “PARADOX”

1.2 Dynamics of Epidemiology in an Epidemic

In the entire Epidemiological narrative, it's illogical to proceed about almost anything unless we know exactly who has caught up the disease and who hasn't. So it's quite succinct that one of the most vital element to begin any sort of epidemiological modeling is “*Testing*”. Obviously, the naive approach is to begin testing every single individual in the region and then tinker with ways to curb the viral outbreak. But the irony is that by the time we reach to test the 800th person we might find the 1st person to whom when we tested earlier in the beginning and tested negative, has been caught with the disease. So all efforts got wasted? Should we restart the entire testing campaign? or Should we just not care about all this and take action based on whatever data is collected? The need of the hour is not to answer these questions but to thoroughly understand those questions, to understand exactly where all this is going, and understand the subtleties lying underneath the core problem. All the above was just regarding one such typical problem that might be encountered while going about Testing, there could be tons of other problems waiting to be questioned and solved. A principled approach to testing is paramount and should be regarded as the highest priority task at hand, in the World of epidemiology.

1.3 BMC - Measles Dataset

Work done over “Testing”, presented herein, is in the Measles Disease spread context, for which the Dataset comes from the Brihanmumbai Municipal Corporation “BMC”, below is a complete breakdown of all the possibly relevant information that could be deduced from the corpus.

Age Bin	Average	Min	Max	Count
0-1	0.63	0	0.92	349
1-2	1.38	1	1.92	238
2-3	2.37	2	2.92	131
3-4	3.42	3	3.92	88
4-5	4.43	4	4.92	90
5-6	5.33	5	5.92	64
6-7	6.42	6	6.92	53
7-8	7.35	7	7.92	62
8-9	8.4	8	8.83	37
9-10	9.41	9	9.92	33
10-12	10.81	10	11.92	51
12-16	13.47	12	15.67	30
16-20	17.93	16.08	19.92	20
20-25	21.93	20	24.75	18
25-30	28.17	25.67	29.92	8
30-40	32.125	30.25	35.75	6
40-50	42.73	40.42	47.33	5
50-60	50.63	50	51.25	2
60-70	62.5	60	65	2

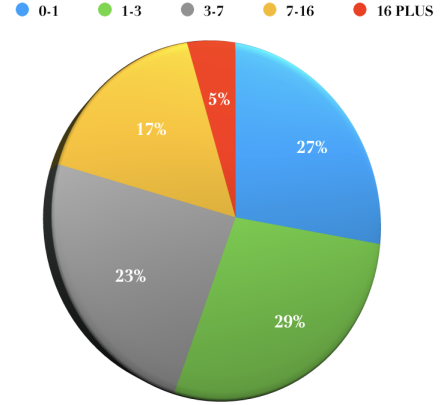


Figure 1.3: Age Wise Statistics

The Pie chart along with the summarized dataset shows the age-wise breakdown of the data inferred from the BMC-provided dataset. It shows there is a huge dominance of infants rather than adults!

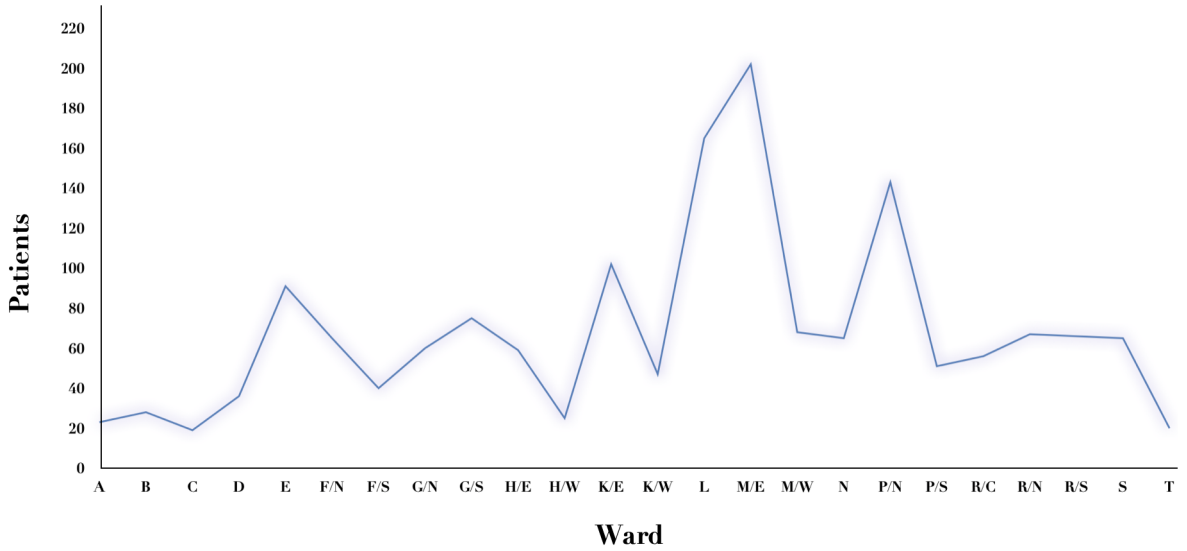


Figure 1.4: Ward Wise Statistics

The Line-Graph shows the Ward-Wise Statistics inferred from the BMC dataset. The above graph lists the concentration of the number of patients across

the wards in Mumbai.

The Skewness of the Graph, quite succinctly speaks about the higher concentration of infection spread in various regions. Wards listed on the X-axis are actually consecutively located in Mumbai, thus very elegantly one can infer the pattern of the Infection Spread for adjacent wards across Mumbai City by a mere stare at it.

A very nice “Bell-Shaped” Curve is formed here, indicating the higher concentrations of Infection near Ward L and M/E and thereby the infection intensity just feels like smearing away across wards on either side.

Hospital	Patients
Kasturba Hospital	190
J J Hospital	26
Jijamata Nagar Dispensary	19
Jijamata Health Post	15
Rajawadi Hospital	14
Cooper Hospital	12
Dr. B.R. Ambedkar Hospital	12
Dr. Prashant Gandhi Hospital	12
Dr. Rupesh Mandodkar Hospital	12
Marol Health Post	12
Malvani Health Post	11
D.R. Shinde’s Children Hospital	10
Gov General Hospital	10
M.W. Desai Hospital	10
Mankhurd Health Post	10
Maruti Lane Dispensary	10
Subhash Nagar Health Post	10
Others	892

Figure 1.5: Hospital Wise Statistics

The Table shows the Hospital-wise Statistics inferred from the BMC dataset. The above list shows the number of people who were reported to be infected by each hospital. Listed here are only the Hospitals that had at least 10 people reported as Infected and all other hospitals had less than 10 reported cases and hence categorized as “Others” above.

Graphs seen till now were among the most important aspects one could infer from the dataset, although there are some more modalities that need to be explored from the dataset, like, Pin Code Wise Statistics, Travel Frequency via Public Transports viz. Railways, the BEST Busses, etc, and because the infected population is predominantly dominated by infants and school-going children, a track over the school location could also lead us to deeper insights related to what the actual infection spread hubs are.

Chapter 2

Graph Signal Processing

In this Chapter we will talk in general about Graph Signal Processing and Graph Laplacian.

2.1 Graph Signal Processing

Graph signal processing (GSP) is a field that deals with signals defined on graphs, where graphs are mathematical objects consisting of vertices and edges that model relationships between different entities. In GSP, signals are represented by vectors whose components correspond to values assigned to vertices in a graph.

One of the key challenges in GSP is to develop methods that can handle the high-dimensional nature of signals on graphs. Since graphs can have a large number of vertices and edges, signal processing techniques that are designed for low-dimensional data may not be directly applicable to graphs.

To overcome this challenge, GSP researchers have developed a variety of techniques that leverage the underlying graph structure. For example, the graph Laplacian matrix, which encodes information about the relationships between vertices, has been used to define graph Fourier transforms and associated frequency-domain representations of graph signals. This has allowed researchers to apply tools from classical Fourier analysis to the graph setting, enabling them to develop new filtering, sampling, and compression methods for graph signals.

Another key area of research in GSP is the development of graph learning algorithms, which aim to learn the underlying graph structure from data. This is an important task in many applications, such as social network analysis, where the graph structure may not be known a priori.

Overall, GSP is an exciting and rapidly growing field that has the potential to revolutionize signal processing by providing powerful tools for analyzing and processing signals on graphs.

2.2 Graph Spectral Domain

Spectral graph theory has traditionally focused on the construction, analysis, and manipulation of graphs, rather than signals on graphs. It has proven to be particularly useful in the creation of expander graphs, graph visualization, spectral

clustering, graph coloring, and many other applications in chemistry, physics, and computer science.

In the field of signal processing on graphs, spectral graph theory has been utilized as a tool to define frequency spectra and expansion bases for graph Fourier transforms. In this section, we will provide a brief overview of some fundamental definitions and notations from spectral graph theory, emphasizing how it enables us to extend many of the important mathematical concepts and intuitions from classical Fourier analysis to the graph setting.

2.3 Graph Laplacian

The unnormalized graph Laplacian, also known as the combinatorial graph Laplacian, is defined as:

$$L = D - W \quad (2.1)$$

where D is a diagonal matrix with the i th diagonal element d_i equal to the sum of the weights of all the edges incident to vertex i , and W is the adjacency matrix of the graph.

The Laplacian is a difference operator for any signal f in R^N , and it satisfies:

$$(Lf)(i) = \sum_{j \in N_i} (W_{i,j}[f(i) - f(j)]) \quad (2.2)$$

where N_i is the set of vertices connected to vertex i by an edge, and $N(i, k)$ is the set of vertices connected to vertex i by a path of k or fewer edges.

The graph Laplacian L is a real symmetric matrix with a complete set of orthonormal eigenvectors denoted as $u_0, u_1, \dots, u_{N-1}$. The associated eigenvalues are real and nonnegative, denoted as $\lambda_0, \lambda_1, \dots, \lambda_{N-1}$, satisfying $Lu_i = \lambda_i u_i$ for $i = 0, 1, \dots, (N-1)$. The eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The number of connected components of the graph determines the multiplicity of the eigenvalue 0, and since we consider connected graphs, we assume that the graph Laplacian eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The set of all eigenvalues is denoted as $v(L) = \lambda_0, \lambda_1, \dots, \lambda_{N-1}$.

2.4 Compressed Sensing Theory for Reconstruction of Partially known Signals

Compressed sensing (CS) is a powerful framework for reconstructing unknown signals from sparse measurements. It has been widely used in various signal processing applications, including image processing, signal acquisition, and data compression. In the context of graph signal processing, CS has been applied to reconstruct graph signals from a limited number of measurements.

The basic idea of CS is to leverage sparsity, or compressibility, of signals in some basis or transform domain. The key observation is that many real-world signals can be represented by a sparse or compressible linear combination of basis functions. In other words, most of the signal energy is concentrated in a few basis coefficients, while the rest of the coefficients are close to zero. CS exploits this

sparsity property by acquiring a small number of linear measurements of the signal and then using an appropriate reconstruction algorithm to recover the original signal from the measurements.

In the graph signal processing setting, the graph Laplacian matrix plays a central role in CS-based signal reconstruction. The graph Laplacian encodes the structural information about the graph, including the connectivity and the degree of each vertex. It also provides a natural framework for defining a graph Fourier transform and associated frequency-domain representations of graph signals.

To apply CS to graph signals, one can represent the graph signal in the graph Fourier domain, where the signal is sparse or compressible. The sparse representation can be obtained by solving an optimization problem that minimizes the ℓ_1 -norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. The resulting optimization problem is a convex optimization problem that can be efficiently solved using various algorithms such as iterative soft thresholding (IST) and orthogonal matching pursuit (OMP).

The key challenge in applying CS to graph signals is to design appropriate measurement matrices that allow for accurate signal reconstruction from a small number of measurements. In general, the measurement matrix should be incoherent with the graph Fourier basis and should satisfy certain properties such as the restricted isometry property (RIP) to ensure accurate signal recovery.

In summary, compressed sensing theory has been applied to graph signal processing by leveraging the sparsity property of graph signals in the graph Fourier domain. The graph Laplacian matrix provides a natural framework for defining the graph Fourier transform and associated frequency-domain representations of graph signals. CS-based signal reconstruction can be achieved by solving a convex optimization problem that minimizes the ℓ_1 -norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. Appropriate measurement matrices that are incoherent with the graph Fourier basis and satisfy certain properties such as RIP are essential for accurate signal recovery.

Chapter 3

Graph Signal Processing for Testing

In this chapter we will deeply analyse 2 research papers which have worked in this field, viz., "Graph Frequency Analysis of Covid-19 Incidence to Identify County level Contagion Patterns In the United States" by Li et. al. and "Kemeny-based testing for COVID-19" by Yilmaz et. al.

3.1 Graph Frequency Analysis of Covid-19 Incidence to Identify County level Contagion Patterns In the United States

The study examines the spread patterns of COVID-19 in US counties through a graph frequency analysis. Commute flows between all 3142 US counties were used to construct a graph representing population mobility, and the numbers of daily confirmed cases were collected as graph signals. The graph Fourier transform was used to map the graph signals into the frequency domain, enabling the decomposition of signals into modes with smooth or rapid variations. The study shows that different modes of variability relate to COVID-19 spread patterns within and across counties, revealing changes in the nature of spread within geographical regions at finer temporal scales. The approach demonstrates the potential of using Graph Signal Processing (GSP) tools for epidemiology knowledge discovery on graphs.

Graph representations of networks enable the analysis of complex data from various aspects of human life. Graph Signal Processing (GSP) emerged to leverage graph structures for information processing, and it has been applied to various timely applications. This study uses recent GSP advances to investigate the spread pattern of COVID-19 across all US counties, providing a comprehensive spatio-temporal analysis of the contagion. The study involves spatio-temporal analysis of case data from 3142 US counties, graph frequency analysis using a GFT-based approach, and investigation of changes in the nature of contagion patterns within counties. The results reveal fundamentally different contagion patterns across locations and help identify counties at risk of major outbreaks in a data-driven manner.

3.1.1 GSP fundamentals

The Paper considers signals that are associated with nodes of a weighted, undirected, and connected graph. More formally, we can define the signal vector $x = [x_1, \dots, x_N]^T \in R^N$ and the weighted graph $G(V, E, W)$, where $V = 1, \dots, N$ represents the set of nodes and $E \subseteq V \times V$ is the set of edges. Here, x_i denotes the signal value at node $i \in V$, and the map $W : V \times V \rightarrow R^+$ associates a non-negative weight $W_{ij} \geq 0$ with the edge $(i, j) \in E$, while $W_{ij} \equiv 0$ for $(i, j) \notin E$. The symmetric coefficients $W_{ij} = W_{ji}$ capture the strength of the connection between nodes i and j , reflecting their similarity or influence.

For the purpose of this paper, they focus on a specific graph consisting of $N = 3142$ counties in the United States. The edge weights in this graph correspond to the average amount of population commuting between counties. The signal x represents the daily number of confirmed COVID-19 cases across all counties. One can interpret the edge weights as capturing the strength of the connection between counties, which reflects the potential for virus transmission based on population commuting flows. If the weight W_{ij} is large, the signal values x_i and x_j tend to be similar, indicating a higher likelihood of virus transmission. Conversely, if the weight W_{ij} is small, the signal values x_i and x_j are less directly related, except through their weak connections to other nodes.

This interpretation of the edge weights establishes a link between the signal values and the graph topology, motivating the use of a novel Graph Signal Processing (GSP) approach, which is proposed in this paper.

3.1.2 GFT and Smoothness

GFT which is Graph Fourier transform is an important tool in the Graph Signal Processing World, as it allows to represent the graph signal into 2 different domains:

1. Vertex Domain, consisting of the nodes in V
2. Graph Frequency Domain spanned by the spectral basis of G , Thus to induce the different levels of interaction between the neighbors the signals can be manipulated in the frequency domain.

To expound on this concept, we can use the eigenvector decomposition of the combinatorial graph Laplacian $L := \text{diag}(W_1) - W$ to define the Graph Fourier Transform (GFT) and the associated concept of graph frequencies. We can always decompose the symmetric graph Laplacian as $L = V\Lambda V^T$, where $\Lambda := \text{diag}(\lambda_1, \dots, \lambda_N)$ denotes the diagonal matrix of non-negative Laplacian eigenvalues and $V := [v_1, \dots, v_N]$ is the orthonormal matrix of eigenvectors.

The GFT of a signal x with respect to the Laplacian L is the signal $x_\sim = [x_\sim 1, \dots, x_\sim N]^T$ defined as $x_\sim = V^T x$. The inverse GFT (iGFT) of $x_\sim$ is given by $x = V x_\sim$, which is a proper inverse due to the orthonormality of V .

The formula for iGFT, $x = V x_\sim = \sum_{k=1}^N x_\sim^{(k)} v_k$, allows for the synthesis of x as a sum of orthogonal frequency components v_k , where the contribution of each v_k to the signal x is given by its corresponding GFT coefficient $x_\sim^{(k)}$. The GFT captures the concept of signal variability over the graph, similar to how frequency captures variability in Fourier analysis of temporal signals.

To understand this analogy, we can define the total variation of the graph signal x with respect to the Laplacian L (also known as Dirichlet energy) as the

quadratic form $TV(x) := x^\top Lx = \sum_{i \neq j} W_{ij}(x_i - x_j)^2$. The total variation $TV(x)$ serves as a measure of smoothness, quantifying the amount of change in the signal x with respect to the variability encoded by the weights W .

Returning to the GFT, we can examine the total variation of the eigenvectors v_k , which is given by $TV(v_k) = v_k^\top Lv_k = \lambda_k$. Therefore, the eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$ can be interpreted as graph frequencies, indicating how the eigenvectors (i.e., frequency components) vary over the graph G . As a result, the GFT and iGFT provide a means of decomposing the graph signal x into spectral components that capture different levels of variability.

For a graph signal x with GFT coefficients x , filtering can be performed in the frequency domain similarly to classical signal processing of time-varying signals. Eigenvalues of the Laplacian correspond to graph frequencies and eigenvectors act as a frequency basis. Using this, low-pass, band-pass, and high-pass graph filters can be defined by isolating specific eigenvalues and their corresponding eigenvectors. These filters are mutually exclusive and span all graph frequencies. By mapping signals to the spectral domain via the GFT, filtering them, and then mapping each frequency component back to the vertex domain via the iGFT, the original graph signal can be decomposed into components exhibiting low, medium, and high variability with respect to the graph. This decomposition increases the resolution of the signal and can be used to investigate different patterns across the graph.

3.1.3 GSP and COVID-19

Graph frequency analysis conducted on US COVID-19 prevalence data, defining signals and constructed graph for study, followed by thorough frequency analysis to identify contagion patterns across counties.

COVID-19 data as Graph Signals

The section discusses how COVID-19 data can be represented as graph signals by using the number of daily confirmed cases per 100k residents of each county in the US, and stacking these signals as columns of a matrix. The study takes into account the commute flow between counties in building a network graph, which helps to reveal hidden relationships between signals and identify different patterns of virus contagion across different locations in the country.

Network Graph Construction

The study uses data from the US Census Bureau to construct a weighted undirected graph with 3142 counties as nodes and edges representing population mobility flows between counties, normalized by intra-flows. The resulting graph shows dense diagonal blocks as expected due to mobility within states, but also significant connections between counties in different states, suggesting long distance commutes. This network provides richer pairwise similarity patterns than the distance-based graph used in previous work.

Frequency decomposition of graph signals

In this section, the authors use the graph signals X and graph G from the

previous sections to carry out frequency decomposition of the daily case signals. They form the Laplacian matrix using W and compute its eigenvalues and eigenvectors. A low/high-pass filter is constructed by taking the lowest/highest one fifth of the eigenvalues, respectively. After graph filtering, the original graph signal X is decomposed into XL, XM, XH which represents signal components that change slowly/mildly/rapidly with respect to the underlying commute graph G . The authors focus on low-pass and high-pass signals as they correlate with distinct contagion patterns. They take the row-wise average of the absolute values in XL and XH to obtain two vectors of length 3142 that quantify the signal magnitude per county with respect to their energy occupancy in low-pass and high-pass spectral band. They use LP regions to represent counties with high magnitude of low-pass signals, and HP regions for counties with high magnitude of high-pass signals. They decompose the signal of each county into low-pass and high-pass frequency contributions and assess whether its signal concentrates more in either frequency band. They overlay the magnitudes of low-pass signals and high-pass signals of all counties on a US map and observe that LP regions have similar number of daily confirmed cases normalized by population size with neighboring counties, while HP regions have markedly distinct and abnormal case counts relative to nearby counties, despite the population mobility between them. Analysis in the next section establishes a link between graph frequencies (low/high-pass) and contagion spread patterns based on descriptive analysis of the graph signals.

Frequency Analysis

They analyze the frequency components of the GFT to identify patterns of spread across counties. The authors explain that low-pass signals (smooth, diffusive signals) in the GFT correspond to cross-county contagion patterns, where nearby counties tend to exhibit similar numbers of confirmed cases. On the other hand, high-pass signals (spiky signals) correspond to within-county outbreaks, where the signal of a county is very dissimilar to nearby counties. The authors use these frequency components to identify regions with cross-county spread and within-county outbreaks.

The authors compute several metrics for these regions, such as average daily cases and the number of days to reach peaks. They find that LP regions have more daily cases and take longer to reach peaks, while HP regions have fewer daily cases and reach peaks faster. They argue that LP regions suffer from cross-county spread with a continuous increasing rate and a longer contagion period, while HP regions suffer from local outbreaks in highly concentrated places.

Finally, the authors analyze the temporal evolution of COVID-19 cases in the New York City area, using weekly periods. They show how a localized outbreak in one county transforms into a wider spread across the region, as seen in the dominance of low-pass signals in the GFT. The authors argue that their approach provides insights into the spread of COVID-19 across different regions and can inform public health policies.

3.2 Kemeny-based testing for COVID-19

At its core, Authors are discussing the importance of testing, tracking, and tracing in controlling the spread of COVID-19 and safely reopening activities. Contact tracing apps allow for the reconstruction of daily contact graphs, which can be used to identify bridges between different communities. The authors propose a new metric called the "Kemeny indicator," which is based on the Kemeny constant of a graph and can be used to identify individuals who are likely to be "super-spreaders" or links between different communities. By testing individuals associated with high Kemeny indicators, early identification and interception of new virus outbreaks can occur. The effectiveness of this approach has been demonstrated through extensive simulations.

Kemeney indicator for criticality

Authors describes a method for identifying the most critical nodes in a graph, which are those that connect different communities and could potentially act as "super-spreaders" in the transmission of disease. This method involves calculating the Kemeny constant of the graph after removing each node and using this value as a measure of the node's importance. Nodes with high Kemeny indicators are those that, if removed, would cause a significant increase in walking times between communities, indicating their criticality.

PageRank and Centrality indicator

Here Authors discuss different measures of centrality in graphs, specifically focusing on PageRank and betweenness centrality indicators. It explains how PageRank corresponds to the Perron eigenvector, and that teleportation is a factor in the calculation. The passage also describes the use of Random Walk Betweenness (RWB) as a measure of betweenness centrality, which is shown to be effective in identifying nodes that bridge communities. The Kemeny constant is also introduced as a measure of the average time to take a random walk in the contagion graph, weighted according to likely destinations, representing a compromise between indicators that use only the stationary distribution and those using path-based algorithms. The passage concludes that the Kemeny constant should work in highly connected single-community graphs, as well as in more sparse graphs associated with many unknown sub-communities. The article presents a framework for using the change in the Kemeny constant of a graph to identify and analyse bridges between sub-communities in order to efficiently identify and block new virus outbreaks as early as possible by removing possible 'super-spreader' links that transmit disease between different communities. The Kemeny constant is a new method used in sampling problems that is related to other methods in the literature, such as the random walk betweenness indicator. Advantages of the Kemeny constant include its efficient computation, its ability to take into account weighted graphs and variable transmission probabilities, and its ability to handle directed graphs with asymmetric transmission paths.

Overall Analysis of Paper

This Article describes a simulation study that used the Kemeny indicator to identify "bridge nodes" (individuals who connect different communities) in a network of interactions. In the first simulation, the graph representing interactions was drawn randomly each day, but in the second simulation, the graph had a fixed structure and random interactions were simulated on the edges. The authors suggest that in real-life scenarios, testing or intervening with these bridge nodes proactively before infection (e.g. through quarantine measures or targeted education/enforcement) may be more effective than testing them repeatedly. They also note that there are privacy concerns about centralized government knowledge of social graph structures, and suggest that distributed estimation of the Kemeny constant may be possible using distributed ledger technology. Finally, they mention that their work builds on a previous paper from 2011, in which the Kemeny indicator was used to identify critical nodes and edges in networks. The authors believe that their work can be extended in a number of ways, such as taking into account: dynamic evolution of contact graphs, using optimized combinations of indicators, investigating approximated distributed approaches for computing the Kemeny constant, using more sophisticated epidemiological models, and applying the findings to immunization procedures against other seasonal diseases.

Chapter 4

Graph Signal Processing for Subset Selection with Contact Tracing Graph

4.1 Nomenclature And Algorithm

$\mathcal{N}$	Total Population = $\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{\mathcal{N}}\}$ where $\mathcal{P}_i$ represents i^{th} Person in the Population. $\mathcal{N}$ is size of Population set.	Scalar
f	Initially infected Population	Scalar
K	Number of days to Simulate	Scalar
ρ	Regularisation Parameter in Convex optimisation problem.	Scalar
τ	Threshold	Scalar
λ	Parameter used in Probability of i^{th} person getting infected from the j^{th} person	Scalar
ViralLoad	Viral Infection Score assigned to every person.	$\mathcal{N} \times 1$
A	Adjacency Matrix. $A[i][j] = 1$ if i and j came in contact $A[i][j] = 0$ if No contact between i^{th} and j^{th} person.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.
T	Time of Duration of Contact between i^{th} and j^{th} person. Random Values assigned.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.
S	Signal Strength between i^{th} and j^{th} person. Inversely proportional to the distance between i^{th} and j^{th} person. Random Values assigned.	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Matrix.

Figure 4.1: Nomenclature Part 1

$\mathcal{P}$	$P[i][j] = 1 - e^{-\lambda * T[i][j] * S[i][j] * \max(\text{ViralLoad}[i], \text{ViralLoad}[j])}$ viral load heuristic School, College, Places in Rail Roads, Frequency of transfer of people from that pg	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix,
D	Row wise sum stored at Diagonal Positions. $D[i][j] = \begin{cases} 0 & \text{if } i \neq j \\ \sum_{j=1}^N P[i][j] & \text{otherwise} \end{cases}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Diagonal Matrix
$\mathcal{L}$	Laplacian Matrix, $\mathcal{L} = D - \mathcal{P}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix, Symmetric Positive Semi Definite Matrix.
V	Eigen Vectors of $\mathcal{L}$	$\mathcal{N} \times \mathcal{N}$ Dimensional Matrix,
X	Binary vector containing health status People. $X[i]=1$ if infected 0 otherwise	$\mathcal{N} \times 1$ Binary Vector

Figure 4.2: Nomenclature Part 2

θ	Our Conjecture, $X = V * \theta$ [Comes from Compressed Sensing Theory] $\theta = V^{-1} * X$ Because V is Orthonormal set of Eigen vectors, $\theta = V^T * X$	$\mathcal{N} \times 1$ Vector
Y	Subset of X	$m \times 1$ Binary Vector $m < \mathcal{N}$
<i>random_indices</i>	Indices of subset of values of X chosen to be stored in Y. If $Y[i] = X[k] \Rightarrow \text{random_indices}[i]=k$	$m \times 1$ vector
ϕ	Row wise one hot encoded Matrix corresponding to Y. for $i = 1: m$ $\phi[i][\text{random_index}[i]]=1$ otherwise 0	$m \times \mathcal{N}$

Figure 4.3: Nomenclature Part 3

1. Set the Hyper-parameters.

$\mathcal{N}$ The total Population Size.

f Initial Infected set of People.

K Total Number of Days to Simulate.

2. Introducing the Community Structure.

Generate the clique structure. For Example, if the total population is $\{P_1, P_2, P_3, \dots P_{15}\}$ then the cliques generated will be $C_1=\{P_1\dots P_6\}$, $C_2=\{P_5\dots P_9\}$, $C_3=\{P_{10}, P_{11}\}$, $C_4=\{P_{11}\dots P_{15}\}$

I have set the overlap factor between any two consecutive cliques to be 50%

3. Constructing the Initial contact tracing matrix A.

This is basically Adjacency Matrix. Now while filling the adjacency matrix, we will strongly connect individuals of each Clique to every other individual of that clique.

$N = 15$

A Adjacency Matrix

There are 3 cliques are here, $C_1 = \{1..5\}$, $C_2 = \{3..5\}$, $C_3 = \{4...15\}$.

0	1	1	1	1	0	0	0	0	0	0	0	0	0	0
1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
1	1	0	1	1	0	0	0	0	0	0	0	0	0	0
1	1	1	0	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	0	1	1	1	1	1	1	1	1	1	1
0	0	0	1	1	0	1	1	1	1	1	1	1	1	1
0	0	0	1	1	1	0	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	0	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	0	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1	0	1	1	1	1
0	0	0	1	1	1	1	1	1	1	1	0	1	1	1
0	0	0	1	1	1	1	1	1	1	1	1	0	1	1
0	0	0	1	1	1	1	1	1	1	1	1	1	0	1
0	0	0	1	1	1	1	1	1	1	1	1	1	1	0

Figure 4.4: Algorithm Part 1

4. Setting the Viral Loads and the Infection Dates of all the individuals.

Randomly assigned Infection scores to everybody. Infection dates to initial set of people set to the day 0, and infection dates of every other person set to $K + 1$, indicating they don't have any infection right now.

5. Begin the Simulation Process for K days.

5.1 Generating the Contact Tracing Map A

Infusing some random contacts like wildfire in the Matrix A.

5.2 Constructing the Probability Matrix P using the Time of Duration of Contact , the Signal Strength of Contact and the Viral Load carried by the people who have been in contact.

If $A[i][j] = 1$ Assign a probability score.
--

5.3 Compute the Eigen Vector Matrix of Laplacian Matrix L of P

$V = \text{Eigen_vector}[L[P]]$

The above are going to Orthonormal set of Eigen vectors, because L is going to be symmetric positive semi definite matrix.

5.4 Diffusion Process

Now simulating the spread of infection by utilising the Contact Tracing Map and the Probabilities computed.

5.5 Compute $\theta_0 = V^{-1} * X$

5.6 Construct the Y Vector.

Y is a subset of X of size $m \times 1$ where $m < \mathcal{N}$

5.7 Construct the ϕ Vector.

Row wise one hot encoding of Y vector.

Figure 4.5: Algorithm Part 2

X vector represents the binary health status of all people in the population	..N x 1
theta vector represents the sparse view of X	..N x 1
V encodes the Transition Matrix and the laplacian	..N x N
V*theta represents the predicted status for all people in population	..N x 1
phi is used to select the predicted status for subset selected y	..m x N
phi*V*theta gives status for subset Y	..m x 1

Figure 4.6: Algorithm Part 3

5.8 Solve the Convex optimisation Problem

$\arg \min_{\theta} \ y - \varphi \hat{V} \theta\ _2^2 + \rho \ \theta\ $
Constraint $0 \leq V^* \theta \leq 1$

5.7 We get θ_1

We compute $X = V * \theta_1$ again to get X_1

This represents the final set of population to select as a subset for testing.

Note: Thresholding needs to be done on X_1 to get Binarized version of it.

So we use the hyper parameter τ .

Figure 4.7: Algorithm Part 4

Chapter 5

Results Obtained

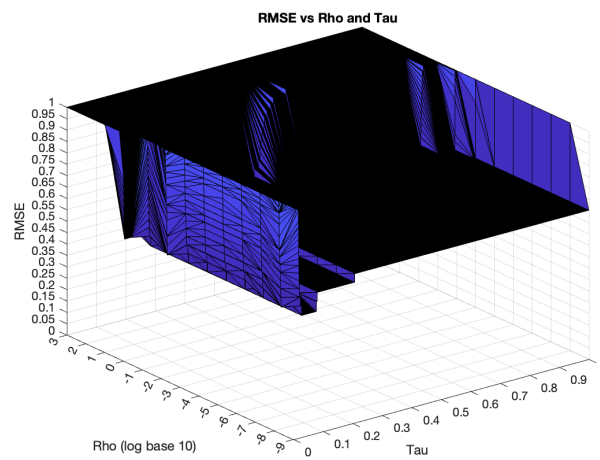


Figure 5.1: RMSE vs Rho and Tau

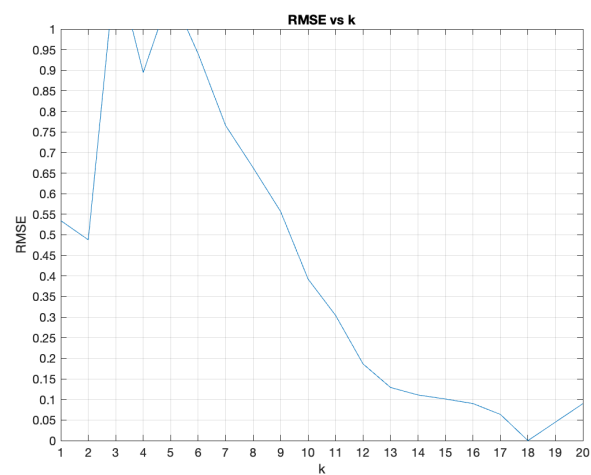


Figure 5.2: RMSE vs Day

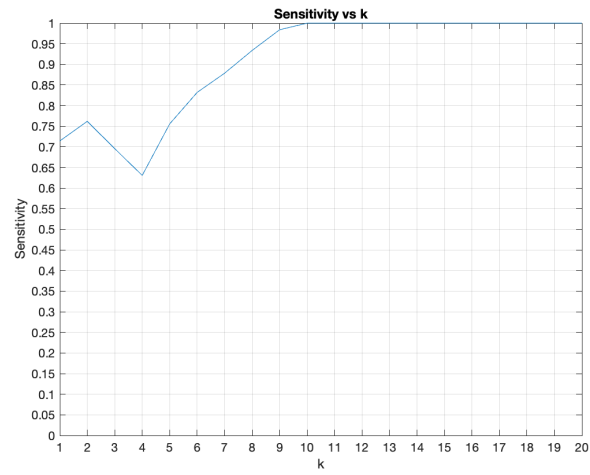


Figure 5.3: Sensitivity vs Day

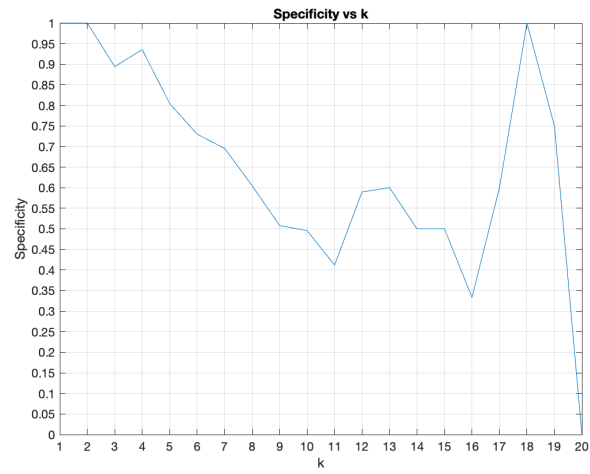


Figure 5.4: Specificity vs Day

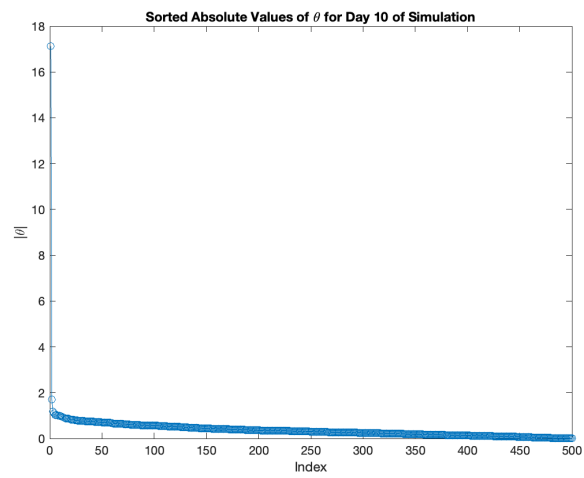


Figure 5.5: Theta plot

Chapter 6

Investigating Graph

6.1 Analysis of Results Obtained

We will discuss 2 aspects, first is the introduction of community structure in the graph and then the various subset selection strategies that could be used.

6.1.1 Necessity of Introducing a Community Structure

Theta plots were supposed to be sparse and they do turn out to be one. But if we look at the other plots, especially the RMSE plots, then it's very clear that the very assumption of building the Contact tracing matrix on an individual level is not the correct choice, and instead a community structure should be introduced and Contact tracing map on community level should be constructed.

6.1.2 Subset Selection Strategies

One more important thing as evident from the algorithm presented is the selection of the subset. There are multiple approaches proposed that we can use to see how contact tracing could be enhanced. Some of the most common strategies are:

1. Node Degree Indicator
2. Page Rank Based Approach
3. Kemeny Based Approach
4. Random Walk Betweenness Approach

The explanation for each of the Subset Selection Strategy:

1. Node Degree Indicator

This approach selects a subset of nodes based on their degree or the number of edges incident on them. The idea behind this approach is that nodes with a high degree are more important because they have more connections to other nodes in the network. This approach is simple and easy to implement, but it may not always produce the best results, as some nodes with a lower degree may still be important in the network.

2. Page Rank Based Approach

This approach uses the concept of PageRank, which is a measure of a node's impor-

tance in a network based on the number and quality of links pointing to it. The idea behind this approach is that nodes that are linked to by other important nodes are also important. PageRank is calculated by iteratively computing a score for each node based on the scores of the nodes that link to it. This approach is widely used in web search engines and other applications.

3. Kemeny Based Approach

The Kemeny constant is a new method used in sampling problems that is related to other methods in the literature, such as the random walk betweenness indicator. Advantages of the Kemeny constant include its efficient computation, its ability to take into account weighted graphs and variable transmission probabilities, and its ability to handle directed graphs with asymmetric transmission paths. This text describes a simulation study that used the Kemeny indicator to identify "bridge nodes" (individuals who connect different communities) in a network of interactions. In the first simulation, the graph representing interactions was drawn randomly each day, but in the second simulation, the graph had a fixed structure and random interactions were simulated on the edges. The authors suggest that in real-life scenarios, testing or intervening with these bridge nodes proactively before infection (e.g. through quarantine measures or targeted education/enforcement) may be more effective than testing them repeatedly. They also note that there are privacy concerns about centralized government knowledge of social graph structures, and suggest that distributed estimation of the Kemeny constant may be possible using distributed ledger technology.

4. Random Walk Betweenness Approach

This approach is based on the concept of random walks on a network. The idea is that nodes with a high random walk betweenness centrality are more important because they are more likely to be visited by a random walk that starts from one node and ends at another. The random walk betweenness centrality of a node is the fraction of random walks that pass through that node. This approach is useful for finding nodes that act as bridges between different parts of a network.